

Apurve Karanwal

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Education

KIET Group of Institutions

B.Tech in Information Technology | Focus: DSA, Systems Architecture, Web3

Ghaziabad, UP

Open Source

Contributions | *Apple, json-schema-org, kgateway-dev*

apurve.xyz/contributions.html →

- Contributed bug fixes, performance optimizations, and documentation updates across major Apple (*coremltools*, *embedding-atlas*), Kubernetes API/AI Gateway (*kgateway*), and *json-schema* ecosystems. Detailed changelogs and pull request links: apurve.xyz/contributions.html.

Projects

PNGProtect | *FastAPI, PyTorch, Solidity, OpenCV, Web3.py, NumPy, Cryptography*

[GitHub](#) | [Live](#)

- Engineered an image security platform using **PyTorch** (FGSM) and **FastAPI** to inject adversarial perturbations, successfully neutralizing feature extraction in vision architectures (CLIP, ResNet, ViT) to prevent unauthorized AI model training.
- Developed a decentralized provenance registry integrating **LSB steganography**, SHA-256/Blake2 cryptographic hashing, and a Solidity smart contract registry deployed on **Ethereum/Polygon** EVM layers via **Web3.py**.
- Built an AI Confusion Visualizer mapping KL-divergence heatmaps, backed by Pillow/NumPy sanitizers that strip device fingerprints, EXIF profiles, GPS coordinates, and device signatures to ensure total data privacy.
- Designed privacy-first metadata sanitizers to completely strip EXIF device profiles, camera serial numbers, and GPS coordinate tracking.
- Implemented an image forensic engine using **OpenCV** and **SciPy** to analyze binary similarity, identify JPEG/PNG compression artifacts, and calculate anomaly scores for tamper detection.

mdfmt | *React 19, TypeScript, Express, Groq, TipTap, Zustand, Prisma, Firebase*

[GitHub](#) | [Live](#)

- Developed a visual Markdown editor using **React 19** and a custom headless **TipTap** (**ProseMirror**) editor with unidirectional data flow bound to a **Zustand** draft persistence store and local storage wrappers.
- Engineered a real-time HTML-to-GFM compiler extending the **Turndown** parser with custom AST node handlers to compile pipe tables, task checklists, inline code snippets, and markdown Alert blocks.
- Built an asynchronous AI Documentation Agent (**Llama 3.3-70B** via **Groq**) with heuristic priority file trees, Mermaid.js Diagram Studio mapping, Firebase OAuth authentication, and database sync via **Prisma ORM**.
- Implemented secure OAuth (Google/GitHub) authentication flows with session persistence and database integration via **Prisma ORM**.
- Created a live Diagram Studio translating natural language schemas into interactive system diagrams rendered in real-time with **Mermaid.js** and synchronized theme states.

PrepFlow AI | *FastAPI, Next.js, WebSockets, Playwright, Groq, Whisper, Neon PostgreSQL* [GitHub](#) | [Live](#)

- Engineered a low-latency voice mock simulator using **FastAPI** **WebSockets** and **Next.js**, streaming low-latency audio via **Whisper-large-v3**, **Sarvam AI** (*saaras:v3* STT, *bulbul:v3* TTS), and OpenAI TTS with sound-wave interrupt detection.
- Developed a mathematical matching and evaluation engine from scratch using custom **TF-IDF** vectors and logarithmic-smoothed IDF cosine similarity equations to grade transcript semantic alignment and keyword coverage.
- Created Greenhouse/Ashby scraping pipelines and a custom **Playwright** browser automation engine to auto-fill applications with LLM-generated context-aware responses, persisting data securely to Neon Postgres.
- Architected a database layer on **Neon Postgres** utilizing serverless scaling and metadata-driven schema migrations, mapping raw query rows to dynamic dictionary wrappers.
- Implemented a background evaluation engine to track candidate verbal metrics, monitoring speech speed (WPM) and filler word frequency to generate live constructive coaching tips.

Technical Skills

Languages: Python, TypeScript, JavaScript, Solidity, Go, SQL, HTML5/CSS3, C/C++

AI Agentic Systems: LLM Orchestration (LlamaIndex, LangChain), Agentic Workflows, PyTorch, Whisper, Vector Search (pgvector), RAG (Retrieval-Augmented Generation), OpenCV, NumPy

Backend Systems: FastAPI, Node.js, Express.js, WebSockets, Web3.py, RESTful APIs, Cryptography, Distributed Systems

Frontend Web: React (19), Next.js (16/Turbopack), Tailwind CSS, TipTap (ProseMirror), Turndown, State Management (Zustand)

Web3 Blockchain: Smart Contracts (Solidity), EVM-compatible Runtimes, Ethers.js, Decentralized Infrastructure

Databases DevOps: PostgreSQL (Neon Serverless), SQLite, Prisma ORM, Redis, Playwright (Automation), Docker, CI/CD (GitHub Actions), Git